

Dương Bảo Duy

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Summary

Cloud/DevOps Engineer with hands-on Infrastructure as Code experience using Terraform, Docker, and GitHub Actions CI/CD, backed by 9 years of software engineering across backend, mobile, and Linux infrastructure. Design and operate a complete multi-environment platform (dev/staging/production) fully defined in Terraform — provisioning containers, VMs, a k3s Kubernetes cluster, and Ceph distributed storage — with Ansible for configuration management. Fast, self-driven learner with a proven record of mastering complex infrastructure independently, with strong IaC fundamentals that transfer across cloud platforms. Comfortable working with technical documentation in English.

Work Experience

Freelance Fullstack Developer & Infrastructure Engineer | Bến Tre, Vietnam | 12/2022 – Present

- Design and operate a multi-environment Infrastructure as Code platform using Terraform and Ansible — provisioning containers, VMs, a k3s Kubernetes cluster, and Ceph storage pools across dev/staging/test/production layers.
- Built and maintain GitHub Actions CI/CD pipelines: automated build-and-deploy workflows and an end-to-end test pipeline with PostgreSQL service containers and Playwright.
- Containerized multiple applications with Docker: authored Dockerfiles and compose-based multi-service deployments; administered Linux servers, overlay networking (Nebula), and reverse proxies.
- Accelerate infrastructure work with AI coding agents: writing and reviewing Terraform/Ansible changes, remote server operations, and deployment troubleshooting.

Senior Software Engineer | Rever Corp | 06/2019 – 12/2022

- Developed and optimized the Rever mobile application with Flutter and backend API integration.
- Improved app performance through refactoring, database query optimization, caching, and practical profiling.
- Implemented automated tests and improved code quality for mobile features used in a production real estate product.
- Integrated REST APIs for data synchronization between mobile clients and backend services.

Senior Software Engineer | VNG Corp | 08/2014 – 05/2019

- Built mobile video call features using WebRTC and native iOS/Android technologies.
- Designed and implemented call quality analysis systems for monitoring real-time communication metrics.
- Developed data processing and backend support tools with Python and Java.

Mobile Developer | BraveSoft Corp | 05/2013 – 07/2014

- Developed Android and iOS applications for client projects.
- Used Android NDK and FFmpeg for low-level media processing features.

Selected Projects

Homelab Infrastructure Platform (Terraform / Ansible / k3s / Ceph)

- Designed a layered Infrastructure as Code architecture (layer0/1/2, each with dev/staging/test/production environments) using Terraform.
- Provisioned containers and VMs via Incus and libvirt providers; managed configuration with the Ansible provider.
- Built a k3s Kubernetes cluster and Ceph distributed storage pool, fully defined and reproducible through Terraform code.
- Implemented Nebula overlay networking, HAProxy/Nginx/Caddy reverse proxies, and NAS-backed storage.

py-agent — Multi-Interface AI Agent Framework

- Built an AI agent framework (CLI, Web, Telegram) with a plugin system, approval workflows, async tool execution, and PostgreSQL-backed persistence.
- Deploy and operate the service on self-managed Linux infrastructure with containerized dependencies.

Skills

- **Infrastructure as Code:** Terraform, Ansible
- **Containers:** Docker, Dockerfile authoring, Docker Compose, Kubernetes (k3s)
- **CI/CD:** GitHub Actions (build/test/deploy pipelines, service containers)
- **Systems / Networking:** Linux administration, Bash, overlay VPN (Nebula), reverse proxies (HAProxy, Nginx, Caddy), distributed storage (Ceph)
- **Languages:** Python, TypeScript/JavaScript, Go, Java, C/C++
- **AI / Automation:** LLM APIs, AI agent tooling, agent-assisted infrastructure operations and automation
- **English:** working proficiency (technical documentation reading/writing)

Education

Bachelor in Mathematics & Computer Science | Ho Chi Minh University of Science | 09/2007 – 05/2013

Certifications

- CCNA, University of Science Computer Center